

Eleven-Layer H1 Failure Chain

Layers 1-6: Environment & Model

Layer 1: curr_vfe hardcoded 0.0

$\Delta F = 0$ always



Layer 2: Zero-action replay

W_a blind to actions



Layer 3: Passive corpus

$p(o|a) = p(o)$: no coupling



Layer 4: free_bits = 1.0 ceiling

Prior entropy degenerate



Layer 5: Encoder+prior+W_z collapse

Warm-start instability



Layer 6: GRU posterior bypass

z_t ignored by decoder

Layers 7-11: Actor & Gate

Layer 7: Gradient starvation

Encoder $\rightarrow 0$ by step 100K



Layer 8: $\|h_t\|_2$ action-invariant

Actor stays uniform



Layer 9: Cold-start deadlock

$P(\text{commit}|H=13) \approx 0\%$



Layer 10: Fresh-sample PG

$\log p(\tilde{a}) \perp A_t$: zero grad



Layer 11: Gate metric paradox

p_{95} rewards variance $\neq$ commit

**v11 Fix: IDL + domain-selective
corpus + decoder-z-only + WM-freeze
 $\rightarrow$ H1 PASS (29.72 $\times$)**